



Rapport

– Project *Estate Mind* –

4DS5 – Group 5

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I. Business Understanding

Context

Real estate is a key sector in Tunisia, affecting households, investors and economic actors alike.

At the international level, this sector is undergoing a major transformation driven by PropTech platforms based on data and artificial intelligence.

In Tunisia, such approaches remain limited, which creates a real opportunity for local innovation.

It is within this context that our project takes shape: **Estate Mind**.

Problem Statement & Target Users

The **Tunisian real-estate market** is experiencing **strong growth**, with annual returns exceeding **9%**. However, despite this apparent dynamism, the process of buying, selling, or investing in property remains **complex** and **risky** for many stakeholders.

For **buyers**, the main challenges lie in navigating a large number of heterogeneous listings, assessing their credibility, avoiding overpriced properties, and identifying potential hidden defects or legal issues before committing to a purchase.

Investors face a different but equally critical set of difficulties. They must monitor emerging neighborhoods, analyze market trends and seasonal effects, anticipate future value appreciation, and minimize exposure to risky projects or unprofitable investments. The lack of consolidated and reliable market intelligence makes these strategic decisions particularly uncertain.

Sellers, on the other hand, often struggle to determine a fair and competitive price for their assets. Conflicting valuations, limited access to comparable transactions, and concerns about regulatory compliance can delay transactions and reduce market efficiency.

These difficulties are further amplified by several structural characteristics of the Tunisian market, including fragmented and heterogeneous data sources, strong regional disparities in pricing, the coexistence of transactions in dinars and foreign currencies, pronounced tourism-driven seasonality, and a complex regulatory framework related to land ownership and foreign investment.

In this context, the central question of the project can be formulated as follows:

How can we optimize real estate decisions in a fragmented and regulated market?

BO & DSO

BO (Business Objective)	DSO (Data Science Objective)
BO1 — Improve Market Reliability Increase transparency and quality of real-estate listings.	DSO1: Automate multi-source data ingestion.
	DSO2: Structure and clean listings using NLP/LLMs.
	DSO3: Track, version, and orchestrate data pipelines.
BO2 — Understand Territorial Dynamics Identify spatio-temporal trends and emerging areas.	DSO1: Model market activity with time-series methods.
	DSO2: Perform spatial aggregation and mapping of listings.
	DSO3: Detect emerging zones and trigger alerts.
BO3 — Estimate Prices with Confidence Deliver accurate and explainable property valuations.	DSO1: Predict prices using supervised learning models.
	DSO2: Integrate temporal and territorial drivers.
	DSO3: Explain predictions using XAI techniques.
	DSO4: Detect under- and over-valued properties.
	DSO5: Expose valuations and explanations through APIs and visual analytics.

BO4 — Support Investment Decisions Optimize choices through personalized analytics.	DSO1: Build user profiles from preferences and behavior.
	DSO2: Simulate profitability and “what-if” scenarios.
	DSO3: Recommend properties and strategies via scoring models.
BO5 — Secure Legal Compliance Reduce regulatory risk and automate conformity checks.	DSO1: Model legal rules in a regulatory knowledge graph.
	DSO2: Continuously update regulatory data.
	DSO3: Compute an automated legal-risk score.
	DSO4: Detect non-compliance and issue alerts.
	DSO5: Orchestrate specialized agents for continuous monitoring.
BO6 — Make the Platform Operational Enable access through AI assistants and automated reporting.	DSO1: Allow interaction via a conversational assistant.
	DSO2: Automatically generate reports and summaries.
	DSO3: Ensure explainability and traceability of outputs.

II. Data Acquisition & Understanding

III. Modeling

IV. Deployment